

## How do personality, team processes and task characteristics relate to job satisfaction and software quality?

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### ABSTRACT

This article analyses the relationships between *personality*, *team processes*, *task characteristics*, *product quality* and *satisfaction* in software development teams. The data analysed here were gathered from a sample of 35 teams of students (105 participants). These teams applied an adaptation of an agile methodology, eXtreme Programming (XP), to develop a software product. We found that the teams with the *highest job satisfaction* are precisely the ones whose members score highest for the personality factors *agreeableness* and *conscientiousness*. The *satisfaction* levels are also higher when the members can decide how to develop and organize their work. On the other hand, the level of *satisfaction* and *cohesion* drops the more *conflict* there is between the team members. Finally, the teams exhibit a significant positive correlation between the personality factor *extraversion* and software product *quality*.

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### 1. Introduction

People are a fundamental and critical concern in software development success or failure. There is research that takes this aspect into account and incorporates people into the software process [39,47,29,53,1]. This is done by analysing people individually and establishing relationships to the activities performed within the project. Then again there is agreement on the fact that people work together to perform interdependent development tasks, and these group interrelationships are complex. This makes team building in software development an important field of study. Its importance lies in the fact that if we can find out what factors lead to higher development team performance, software managers will be able to use this knowledge to build teams.

This paper is about teams and the collective effort of individuals working together towards the common goal of developing a software system. A team is defined as “a number of individuals brought together for a certain task, goal or objective, engaged in frequent face-to-face interaction to execute a task, while the individuals are (mutually) interdependent on each other with regard to the outcome of the task and its execution” [28].

Social psychology is researching team building and considering a series of factors that have been found to affect team performance.

This research has traditionally analysed team member personality factors and their relationship to task characteristics [37,4,23]. But it is accepted that team performance cannot be reliably predicted from team personality composition and task characteristics alone, but rather depends on the interactive effects of team behaviour characteristics [3]. Other factors of team behaviour like interactions between people, including conflict, cohesion, cooperation, communication and climate are being analysed. The results of studies on team building depend on the task performed (although there are general trends that appear to affect all tasks equally). Therefore, software development teams need to be examined separately to discover what factors are influential for this particular task and what likenesses these specific results have to the outcomes for other tasks. Our research examines whether software quality and job satisfaction are related to the team members' personality, task characteristics and the level of cohesion and conflict that they perceive during team work. This article analyses the relationships between *personality*, *team processes*, *task characteristics*, *product quality* and *satisfaction* in software development teams.

In team performance literature, the basic team behaviour model is based chiefly on McGrath's input-process-output model [36]. This is a simple but very effective approach to groups. This model starts by evaluating the group inputs, like members, their qualities and characteristics, as well as the elements of the group environment. These input factors are combined and interact to form group processes, like cohesion, conflict, cooperation and norming, etc. In turn, these group processes have an impact on group effectiveness

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(output). We used the model shown in Fig. 1 for our empirical study. This model divides the examined variables into four basic components: people, task, team processes and team effectiveness.

As regards *people*, we evaluate the personality factors that social psychology research suggests are applicable to team operation and results [37]. Personality factors determine personal preferences, opinions, attitudes, values and characteristics. In other words, everyone has a different personality topology, and this is what differentiates that individual from everybody else. Psychological studies have converged on a range of five personality factors. These factors constitute the topology that has come to dominate a great many investigations, including work on team formation. Like these, our experimental study measures the personalities of team members based on five personality factors (usually known as the *Big Five*) personality dimensions [15]):

- Neuroticism is a broad dimension that includes traits like anxiety, moodiness, irritability or frustration.
- Extraversion is a trait of people who take a trusting and enthusiastic view of others, which is associated with being sociable, assertive and talkative.
- Agreeableness is a trait of showing altruistic concern and emotional support towards other people.
- Openness to experience denotes especially inquisitiveness about new ideas, values, feelings and interests.
- Conscientiousness is a trait of perseverant, scrupulous and responsible behaviour.

When we use the term *personality* here, we are referring to the personality of all development team members. We measured team personality by averaging the scores of each team member for each individual personality factor. We used the mean as an aggregation measure of the group data following the indications of studies run by Barry and Stewart [4], Barrick et al. [3] and van Vianen and De Dreu [50]. This paper follows in the research footsteps of Barrick and Mount [2] and Barry and Stewart [4] that related personality to individual and team effectiveness. We examine how team members' personality is related to software product quality and the level of team member satisfaction.

For the *task*, we have assessed the features of the task performed by the teams considering two characteristics, *interdependency* and *autonomy*. They are considered by social psychologists to be the most relevant characteristics of team behaviour, as Mollerman et al. [37] established in their investigations. Although there are other task characteristics, like task routine [23], several research papers [11,21,37] have revealed that the key task character-

istics influencing team results are autonomy and interdependency. Additionally, these task characteristics are closely linked to the agile software development method [5]. Let us look at what task autonomy and interdependency mean:

- Task autonomy refers to the team's perception of how much freedom they have to make decisions about objectives (what), work methods (how), delivery schedules (when), distribution of work among team members (who) [21,11].
- Task interdependency refers to a situation where the process and result of a task affects the process and result of other tasks [37]. A feature of interdependency is reciprocity, meaning that people are mutually interdependent.

With respect to *team processes*, Tuckman [46] singles out, cohesion, conflict, cooperation, communication, etc., as significant group processes. These processes act as mediating variables in team formation. We selected three aspects of group behaviour or team processes (task conflict, social conflict and cohesion), which, according to Yang and Tang [54], are very much related to better or worse team performance. There is a lot to be said for including other group behaviour factors, but, for practical reasons (questionnaire length and model efficiency), we wanted to start by examining of a limited number of aspects. The meaning of cohesion and conflict is as follows:

- Cohesion is defined as "a group property with individual manifestations of feelings of belongingness or attraction to the group" [31].
- Conflict is an opposition or discrepancy between the ideas, beliefs or interests of the team members. Discussions and disagreements are a usual thing when several people work together and have to explain their opinions and defend their standpoints. Research conducted at group level has found that there are two types of conflict [50,27]: task conflict and social conflict. We consider both of these conflict types in our study. There is task conflict when the parties have different viewpoints, ideas or opinions about a decision that has to be taken or a specific task. Social conflict develops when the parties have incompatible tastes, ideas or values, and there is personal animosity between the team members involved.

Team effectiveness refers to the extent to which the team achieves its goals like productivity, delivery time and product or service quality. According to Hackman and Oldham [23] and Gladstein [22], effectiveness also has to do with aspects that are not

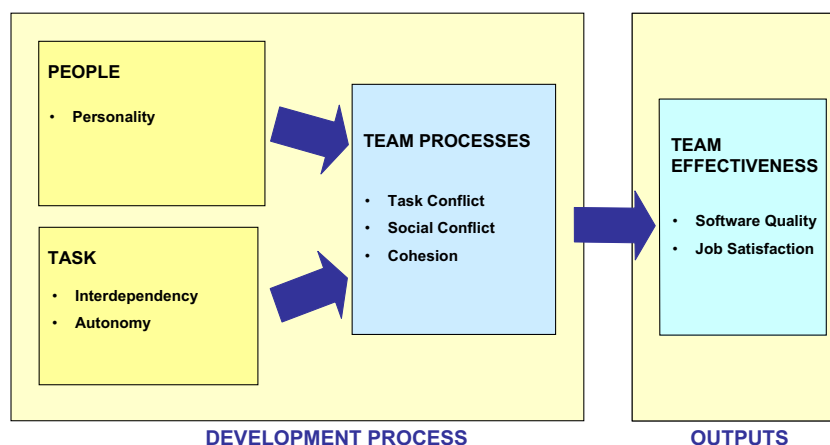


Fig. 1. Modified input-process-output model.

explicitly part of the team's objectives, such as job satisfaction or workability. In research into team effectiveness, the factors usually measured and assessed in the output are satisfaction and performance. Our empirical study examines both these factors:

- Satisfaction indicates how much a team member is in conformity and agreement with his or her team mates regarding the work method, generated atmosphere, objective attainment, etc.
- Developed software quality refers to the appraisal of a set of aspects considered relevant for establishing the level of quality and excellence of the software product developed by its producers (managers, developers, etc.) and receivers (user, customer, etc.).

Research into teams, the quality of the software that they produce and the satisfaction of the members of these teams is becoming increasingly important as more and more software development organizations are moving towards participatory management and looking to reduce hierarchy and organize work on a team basis. Our work analyses the relationships between team *personality*, team *task conflict*, *social conflict* and *cohesion*, task *interdependency* and *autonomy*, and software product *quality* and team *satisfaction* during software development following an adaptation of an agile methodology.

This paper is structured as followed. Section 2 describes related work on team building in the field of social psychology and software engineering. Section 3 details the method of experimental study carried out to relate the personality factors to software quality and team satisfaction, as well as task characteristics and some team processes. Section 4 presents the data analysis and results. Section 5 discusses the results, and Section 6 outlines conclusions.

## 2. Team-related work

In this section, we analyse studies related to team formation and behaviour run in the fields of both social psychology and software engineering. Of these studies, we highlight research dealing with the factors that are most related to our study only, that is, the five personality factors, cohesion, conflict, task characteristics (interdependency and autonomy) and team satisfaction.

### 2.1. Social psychology research on teams

For over 25 years, many researchers have been trying to establish relationships between personality and performance [4,3,40,50,20]. All these papers are based on five personality factors or the *Big Five* personality dimensions: *neuroticism*, *extraversion*, *openness to experience*, *agreeableness* and *conscientiousness*.

Early research looked at these five personality factors in relation to individual performance. Researchers then moved on to consider team performance. For example, Barrick and Mount [2] studied the relationship between personality and individual performance according to three job performance criteria (training proficiency, job proficiency and personnel data) for five occupational groups: professionals, police, managers, sales staff and qualified/semi-qualified workers. They showed, on the one hand, that one of these personality factors, *conscientiousness*, was positively related to individual performance for all the job performance criteria and occupational groups. On the other hand, *extraversion* was also positively correlated to individual performance for management and sales jobs.

Later, Barry and Stewart [4] investigated *extraversion*, *conscientiousness* and intellectual and abstract problem-solving *performance* in relation to group *performance*. These two personality factors were considered at both the individual and team level.

The study was run on a sample of 63 teams of MBA students. They looked at whether the teams benefited when they included a greater proportion of members with the personality traits that were considered important for better operation. Thus, *extraversion* correlated with each team member's individual perception of group performance. It was found that the proportion of extravert members is related to task achievement and team performance. Teams with a moderate proportion of extraverted team members will perform better than teams with few or too many extraverted members. However, the data analysis did not show up any significant relationship to *conscientiousness*.

English et al.'s [20] research analysed the same relationships between team *personality* factors and *performance*, but considered different task types (additive, disjunctive and conjunctive). The additive task considers that the team can successively achieve its objectives by adding together the effort of each team member. The disjunctive task depends on one of the team members being able to do the job successfully. Finally, the conjunctive task requires the input of all the team members, and each team member must meet some minimum requirements. English et al. [20] found that team *conscientiousness* is positively related to team *performance* for additive and disjunctive tasks only. The study participants were 30 three-member cockpit crews from a large international airline carrier.

These studies on the relationships between *personality* and *performance* were extended with the analysis of team processes, i.e. *conflict* and *cohesion*. Specifically, research by Barrick et al. [3] looked at the relationships between:

- Team composition (*general mental ability* and *personality*).
- Team processes (*cohesion* and *conflict*).
- Team results (*workability* and *team performance*).

This study was run with employees from four different companies, although the firms were all in a similar type of business: apparatus assemblers, electronic equipment assemblers, maintenance and manufacturing. The sample totalled 51 already formed teams. The task was to manufacture apparatus, and its key characteristic was interdependency. This study measured how the team member *personality* factors influenced team *performance* and *effectiveness*, also analysing social *cohesion* as a possible trigger of team composition characteristics.

The results of the study showed, on the one hand, that team *performance* was rated high for teams with high scores for *general mental ability* (GMA), *conscientiousness*, *agreeableness*, *extraversion* and low scores for *neuroticism*. On the other hand, teams with high GMA and *extraversion* and low *neuroticism* scores did well on cohesion-mediated team workability. Finally, *cohesion*, plus all the personality factors, except *openness to experience* correlated negatively with *conflict*. Teams scoring high on *agreeableness* and *extraversion* correlated positively with *cohesion*, whereas *neuroticism* correlated negatively with *cohesion*.

Based precisely on this research by Barrick et al. [3], van Vianen and De Dreu [50] ran another study analysing the relationships between team member *personality*, and team *cohesion* and *performance*. The study was conducted first on a sample of 24 teams of professionals employed to lay cables and pipes in the subway system and second on a sample of 28 teams of students working on a research project. In both cases, the tasks had additive, conjunctive and disjunctive components. Some of the results were:

- Average to high levels of *extraversion* and low levels of *neuroticism* contribute positively to team *cohesion*.
- Team *performance* can suffer if a lot of team members are autonomous, although this is task (additive, conjunctive or disjunctive) dependent. High levels of *conscientiousness* positively

contribute to team *performance*. Average levels of *agreeableness* have a significant relationship to *performance*. These results suggested that high or average levels of *conscientiousness* and *agreeableness*, respectively, are necessary to improve *performance*.

Neuman and Wright's study [40] evaluated the combination of *cognitive abilities*, *work skills* and *personality* to predict team *performance*. It focused on determining whether *personality* contributes to *performance* beyond individual and group *skills* and *abilities*. The study was conducted on a sample of 79 four-person teams. They were employees of a wholesale warehouse outlet. The teams were organized to maximize member interaction and interdependency. The results of the study included:

- *Agreeableness* and *conscientiousness* proved to be predictors of *performance*.
- *Agreeableness* is negatively related to *conflict* and positively related to *cohesion*.

Molleman et al. [37] are the only researchers to study the mediational role of task *autonomy* on the relationship between three personality traits (*conscientiousness*, *neuroticism* and *openness to experience*) and two individual results (*satisfaction* and *learning*). The study was conducted on 133 groups of undergraduate business students. The main purpose of the course was to apply behavioural and design theories to describe and analyse organizational problems. The teams had different levels of task autonomy. The multi-level analysis showed that:

- Team task *autonomy* strengthens the relationship between *conscientiousness* and *learning*.
- Team task *autonomy* strengthens the relationship between *openness to experience* and *satisfaction*.

Our study picks up the baton from all the above investigations and sets out to analyse what relationships there are between team member *personality* factors, on the one hand, and their levels of *satisfaction* and the *quality* of the developed software product, on the other. The software development process is composed of conjunctive tasks, that is, all the team members provide input for and are necessary to carry out the development activities and build the software products. Social psychologists have analysed and found other aspects to be related with *satisfaction* and software product *quality*. These factors should also be taken into account in development teamwork. For this reason, this paper also examines relationships with and between team *cohesion* and *conflict* processes, and task characteristics, like *autonomy* and *interdependency*.

## 2.2. Research on software engineering teams

Citing DeMarco and Lister [18], "Most software development projects fail because of failures with the team running them". In actual fact there is widespread recognition that software process productivity and efficiency is critically dependent on human and social factors [9]. A number of studies have been conducted on how to bring the people perspective into software engineering research.

Most of the early research was conducted at the "micro" level, that is, at the individual level with developers involved in programming in the small. This research focused on individual programmer demands and aspects such as the role played by the differences between novice/expert programmers on determining the software project results. Latterly, research has been developed at the "macro" level, that is, at the team and organizational level of people working together on large-scale programming within a software developer organization [16]. This research has dealt with

the role of team and organizational factors within the software process [17,43].

There has been little research into group aspects applied to software development. There are studies that use a standard test, like the "Myers-Briggs Type Indicator" [42,10,45,24], to determine the guidelines for team success according to software engineer personality types. There is another study [52] that determines the connection between abilities and personality traits and team performance. This study was conducted on highly consolidated teams of professionals, and the key factor that it examined was the routineness or non-routineness of the tasks to be carried out.

There are also quantitative team forming methods based on a quantitative abilities model [12], but they do not consider the softer aspects, such as team member personality factors. Another team forming method is based on analysing required and available skills [55], but it does not indicate how to evaluate people's skills. Neither does it consider task and group characteristics.

Zuser and Grechening [56] propose the use of a questionnaire based on abilities and personality traits that provide the team with information during development on finished software projects to improve team performance. The team is built according to the team forming phases of Tuckman's model: forming, storming, norming, performing and adjourning [46].

However, there are few papers on experimental studies dealing with team processes. A study by Yang and Tang [54] examined social *cohesion* and *conflict* team processes and software system development *performance*, considering the interactions between all the development team members, ranging from users, through other roles (programmers, analysts, etc.) to project managers. The findings were:

- Group *cohesion* is positively related to *performance*.
- Group *conflict* is not related to team *performance*.
- *Cohesion* and *conflict* rose and fell depending on the project phases.

Of the agile methods, XP is described as "...a software development method that regards people, rather than paper, as a project's most potent element" [34]. There is abundant literature on experiences with XP processes that support XP's effectiveness, albeit applying a selected subset rather than all its principles and practices [48] and focusing on aspects like confidence, program quality, and team satisfaction to determine the results of small- to medium-sized software projects [35].

Other papers examined developer job satisfaction or product quality after applying certain practices, specifically XP. These papers do not aim to relate either satisfaction or quality to the factors that we examine (personality, team processes and task characteristics). Even so, as they looked at how XP influences job satisfaction and quality, they are to some extent related to our work. Mannaro et al. [33] wanted to find out how important job satisfaction was for the effectiveness of the software development process. To do this, they conducted a survey comparing the job satisfaction of developers using XP practices with developers using other methods. The results of their survey indicated that developers took a favourable view of XP practices and developers using XP practices were more at home with their job environment and better satisfied with their jobs than the others. Macias et al. [32] surveyed 10 XP teams and 10 teams using traditional development practices to measure the internal and external quality of the products they developed. Unlike Wellington et al. [51], they did not find any between-team difference in internal or external quality.

In a systematic review of empirical studies on agile software development [19], the studies were grouped under four headings: introduction and adoption, human and social factors, perceptions of agile methods, and comparative studies. These articles look at

the pros and cons of applying agile methods *per se* or compared against other heavier methods and what factors (developer team cohesion, job satisfaction, team productivity, etc.) agile methods advance or discourage. But none of the reviewed papers examines the relationships between personality factors, team processes and software product quality or team satisfaction. In our research, we focus on this relationship during the design and development of small- to medium-sized software, where all the teams adopt some XP practices. Knowledge like this can lead to recommendations giving the team manager guidance on how to form better teams depending on the task characteristics and team member personality and by introducing group dynamics to resolve conflicts or improve team cohesion (team processes).

### 2.3. Summary of the state of the research on team formation

Table 1 is a summary of the research described above that examined the possible relationships of the five team member personality factors (*neuroticism*, *extraversion*, *openness to experience*, *agreeableness* and *conscientiousness*) to team processes like *cohesion* and *conflict* and other aspects like team *satisfaction* and *performance*.

The columns represent the team processes and the *performance* and *satisfaction* variables that the studies analysed; the rows indicate the five *personality* and *cohesion* factors. The cells in Table 1 are blank when the studies either did not find or did not examine the relationship. The existing relationships are indicated with a reference to the study preceded by a sign (+/–). The positive sign (+) or negative sign (–) means that there is a positive or negative relationship, respectively, between the *personality* factor, the team process and/or the *performance* and *satisfaction* variables.

Table 1 shows that most studies considered *personality* factors and *performance*, which highlights their importance with regard to team formation and behaviour. But, as mentioned above, it also indicates that some authors have studied other aspects affecting teams, like *satisfaction* and team processes.

On the one hand, the result of comparing the studies examining only *extraversion*, *agreeableness* and *conscientiousness* in relation to team *performance* is that there is a positive relationship between these three team personality factors and performance for different task types. Therefore, these personality factors are critical with respect to team formation and its results. On the other hand, Mollman et al.'s study [37] is the only piece of research to show up the existence of a positive relationship between *openness to experience* and team *satisfaction*. Additionally, this study analyses how an underlying variable like task autonomy mediates in the relationship between *openness to experience* and *satisfaction*.

**Table 1**

Summary of the findings of social psychology and software engineering research on teams

|                        | Cohesion        | Conflict        | Performance                         | Satisfaction                        |
|------------------------|-----------------|-----------------|-------------------------------------|-------------------------------------|
| Conscientiousness      |                 | –[3]            | + [3]<br>+ [40]<br>+ [50]<br>+ [20] |                                     |
| Extraversion           | + [3]<br>+ [50] | – [3]           | + [4]<br>+ [3]                      |                                     |
| Agreeableness          | + [3]<br>+ [40] | – [3]<br>– [40] | + [3]<br>+ [40]<br>+ [50]           |                                     |
| Neuroticism            | – [3]<br>– [50] | + [3]           | – [3]                               |                                     |
| Openness to experience |                 |                 |                                     | + [37] (Task autonomy as moderator) |
| Cohesion               |                 | – [3]           | + [54]                              |                                     |

We are going to study how *satisfaction* is related to all the personality factors, team processes and task characteristics. These relationships will also be analysed considering the *quality* of the software produced by the teams involved in our research. Additionally, we are going to look at whether all these relationships apply in software development teams building software according to an agile methodology.

### 3. Method

The empirical study run fits the description of a quasi-experiment [14]. Quasi-experiments are run when the subjects cannot be randomly assigned to an experimental condition or, alternatively, a treatment cannot be randomly assigned to a group. In other words, the group applies a given treatment, but this treatment is not allocated at random. A quasi-experiment is characterized by not being very intrusive or costly. Statistical tests, like correlations and regressions, are applied to the data collected, as applicable.

In quasi-experiments, however, it is hard to satisfy the conditions required to establish a causal relationship between independent and dependent variables (internal validity). The key potential internal validity threats to a quasi-experiment include [14]: subject familiarization with the assessment tests (repeatedly evaluated subjects are exposed to the effects of practice or just get accustomed to the tests); effects due to the sample selection depending on their characteristics with respect to the independent variable (e.g., selecting subjects based on their personal traits); and non-random loss of subjects from samples (the likelihood of this loss occurring is higher in quasi-experimental research). On the other hand, it is easier for the researcher to generalize the research results to the natural environment where the researched processes take place (external validity) than with experiments, as quasi-experiments respect and mirror more closely the conditions occurring in the natural environment.

This study is designed as correlational research [14]. We now describe the goals, hypotheses, parameters, variables, the participants, the subjects, the measurement instruments, and the quasi-experimental procedure. We also characterize the development project and the internal and external validity threats to the quasi-experiment. This study follows the guidelines for reporting empirical research in software engineering [26] as closely as possible.

#### 3.1. Goals, hypotheses, parameters and variables

The goal of this research is to analyse the relationships between three development characteristics with two output characteristics. The three development characteristics that we study are:

- Big five personality factors (*neuroticism*, *extraversion*, *openness to experience*, *agreeableness* and *conscientiousness*) at team level.
- Task characteristics (autonomy and interdependency).
- Team processes (*cohesion*, *task conflict* and *social conflict*).

The two output characteristics are:

- Quality of the developed software.
- Mean satisfaction of the team members.

The context of the study is defined as a special-purpose project with non-professional participants (undergraduate students) undertaking a (toy) project using an adaptation of the agile XP method within a laboratory environment (on-line).

The alternative hypotheses derived from the objective were:

H<sub>1</sub>: There are relationships between all team member *personality* factors: *neuroticism*, *extraversion*, *openness to experience*, *agreeableness* and *conscientiousness*.

H<sub>2</sub>: There are relationships between all team member *personality* factors and developed software *quality*.

H<sub>3</sub>: There are relationships between all team member *personality* factors and software development team *satisfaction*.

H<sub>4</sub>: There are relationships between the task characteristics, *autonomy* and *interdependency*, and developed software *quality*.

H<sub>5</sub>: There are relationships between the task characteristics, *autonomy* and *interdependency*, and software development team *satisfaction*.

H<sub>6</sub>: There are relationships between the team processes, *task conflict*, *social conflict* and *cohesion*, and developed software *quality*.

H<sub>7</sub>: There are relationships between the team processes, *task conflict*, *social conflict* and *cohesion*, and software development team *satisfaction*.

H<sub>8</sub>: There are relationships between team member *personality* factors and the task characteristics, *autonomy* and *interdependency*.

H<sub>9</sub>: There are relationships between team member *personality* factors and the team processes, *task conflict*, *social conflict* and *cohesion*.

H<sub>10</sub>: There are relationships between the task characteristics, *autonomy* and *interdependency*, and the team processes, *task conflict*, *social conflict* and *cohesion*.

H<sub>11</sub>: There is a relationship between *autonomy* and *interdependency* within the team.

H<sub>12</sub>: There are relationships between *task conflict*, *social conflict* and *cohesion* within the team.

Input and process measures (*personality*, *cohesion*, *task conflict*, *social conflict*, *interdependency*, *autonomy* and *satisfaction*) were collected at the individual level and aggregated to the mean level. The team's professor provided the deliverable ratings, and then we assessed the *quality of the software product* developed by each team.

1. *Personality*. The Spanish version of the NEO FFI test was used to evaluate team member personality [15]. This test is composed of 60 questions that measure the five personality factors based on 12 questions per factor:
  - neuroticism,
  - extraversion,
  - openness to experience,
  - agreeableness,
  - conscientiousness.
2. *Cohesion and conflict*. 13-Item questionnaire for evaluating team processes (cohesion and conflict) taken from the Gross Cohesion Questionnaire [44] and an Intragroup Conflict questionnaire developed by [27]. The Intragroup Conflict questionnaire measures both *task conflict* and *social conflict*.
3. *Interdependency and autonomy*. The questionnaire used to determine the task characteristics (interdependency and autonomy) is composed of 12 items. Task interdependency is determined following van der Vegt et al. [49], whereas task autonomy is measured using Molleman's questionnaire [38].
4. *Satisfaction*. Gladstein's questionnaire was used to evaluate satisfaction. Specifically, team satisfaction was evaluated using three items from [22], indicating how satisfied people claim to be with their team mates, working together, etc. This is the response variable used in the regression analyses.

5. *Software Quality*. For this response variable, we selected a set of standard software engineering criteria, primarily taken from SWEBOK 2004 [25] and integrated according to the following formula: decomposition and modularization, testability (the effort it takes to validate the modified software), functionality (software's capability to do what it should, i.e. characteristics relating to achievement of the basic purpose for which the software is being engineered), reusability (the likelihood of a module being able to be used again to add new functionalities with slight or no modification), programming style and team member participation.

All the personality factors, as well as the task characteristics, team processes and satisfaction are evaluated on Likert scales. Students were told that there are no right or wrong answers and that their responses should be accurate and truthful, taking into account the scoring system (1 = totally disagree, 5 = totally agree).

The questionnaires were administered to all 105 individuals on the course. The result is a separate score for each individual and each variable. As already mentioned, because this research was performed at the team level and the questionnaires were administered to individuals, the data had to be aggregated in order to build the team construct. The intra-class correlation (ICC) index can be used to decide whether data can be aggregated. This index compares the inter-group and the intra-group variance [13,30]. The higher the ICC index, the greater the variance at the individual level attributable to the respective team is. Normally, an ICC of over 0.20 is considered to justify aggregation. In our study, all the aggregate variables are significantly higher than this threshold: neuroticism (0.55), extraversion (0.53), openness to experience (0.54), agreeableness (0.52), and conscientiousness (0.52), cohesion (0.60), task conflict (0.59), social conflict (0.58), interdependency (0.55), and autonomy (0.58) and for the team member satisfaction (0.56) response variable. Therefore, we can generate team-level variables by aggregating the scores of the members of each team. We used the arithmetic mean as a measure for aggregating the data for each variable examined in this study at group level, except for the software quality response variable that was measured directly for the team.

To measure the quality of the software product, we analysed the code and project documents. Also we recorded the progress made in the exercises to quantitatively document each team members' participation. This is how we decided on the grade for each team's development project. Table 2 shows the software product quality evaluation criteria that examiners applied to assess the development project and each team adopted to correct its project, plus the associated metrics.

The criteria listed in Table 2 were evaluated on a 1- to 4-point scale. A score of 1 (poor) means that the software product satisfied up to 25% of the aspects that the ideal solution would for each metric. A score of 2 (acceptable) means that software product satisfied from 26% to 50% of the aspects that the ideal solution would. A score of 3 (good) means that the software product satisfied from 51% to 75% of the aspects that the ideal solution would. A score of 4 (excellent) means that the software product satisfied from 76% to 100% of the aspects that the ideal solution would.

Of the total software product quality grade, 20% assessed participation. The remaining 80% assessed the other criteria. The weighted grade on a scale from 0 to 10 points was calculated according to the following formula:

$$\text{Grade} = (((\text{Modularization} * 2 + \text{Testability} * 2 + \text{Functionality} * 2 + \text{Reusability} * 2 + \text{Style} * 2) / 4) * 0.8) + ((\text{Participation} * 10 / 4) * 0.2).$$

At the start of the subject, a common work plan was established for all the teams. This work plan included specific milestones indi-

**Table 2**  
Software product evaluation criteria

| Criteria [5,8,25,41]               | Metric                                                                                                                                                      | Evaluated items            |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| – Decomposition and modularization | – Number of modules and coupling                                                                                                                            | Code and drafted documents |
| – Testability                      | – Number of defects detected by the automatically executed test case set                                                                                    |                            |
| – Functionality                    | – Number of satisfied requirements                                                                                                                          | Team members               |
| – Reusability                      | – Number of reused modules                                                                                                                                  |                            |
| – Programming style                | – Guidelines defined at the subject website <a href="http://www.ii.uam.es/~sacuna/eda/practica.html#6">http://www.ii.uam.es/~sacuna/eda/practica.html#6</a> |                            |
| – Participation                    | – Checklist of laboratory session observations                                                                                                              |                            |

cating when deliverables were to be submitted. The work plan for deliverables was: Deliverable 1 due in the last week of February, Deliverable 2 due in the last week of March, Deliverable 3 due in the last week of April and Deliverable 4 due in the last week of June. These deliverables were evaluated by the professors, and this evaluation served as feedback for students.

The above formula was applied to all scheduled deliverables, except for the first on which reusability was not evaluated. The formula for calculating the final project grade from the partial deliverables was established as:

Final grade = Deliverable 1 grade \* 0.20 + Deliverable 2 grade \* 0.20 + Deliverable 3 grade \* 0.30 + Deliverable 4 grade \* 0.30.

The independent variables were controlled or blocked by keeping conditions constant, i.e. all the variables that were not manipulated in this quasi-experiment have the same level. These variables were as follows:

- *Participant experience*: undesirable factors that can influence the quality of the software produced. This aspect has two sides: software design experience and procedural programming experience. This variable is controlled by selecting subjects at random. It can be argued that our random group selection led to groups with reasonably balanced average subject experience, as the students in our experiment had a reasonably similar background and ability, and the distribution of student grades was normal. Precisely, we conjecture, in the absence of a better hypothesis, that, as the grades were Gaussian, the experience of the participant students is normally distributed.
- *Project under development*: functionalities that the participants should implement during the experiment. If the teams work on different problems, the results of the teams might not be comparable. This variable is blocked by having all teams work on the same problem or project.
- *Knowledge of XP*: developer productivity can be biased by how familiar they are with development method practices. One of the available sessions was taken up with training all participants in XP generally, and especially the XP practices applied.

### 3.2. Participants and subjects

The participants were second-year computing students at the Universidad Autónoma de Madrid's Higher Technical School of Computing. These students were taking the Data Structures and Algorithms subject during the 2004/2005 academic year. The project set was to design and implement software. Through this project, students were expected to demonstrate their knowledge of how to handle and implement a range of abstract data types using different data structures, as well as an ability to analyse and develop software solutions for the stated problem.

A total of 105 students participated in this empirical study, of which 83 were male (80%) and 22 were female (20%). Of the students, 75% (79 students) were aged less than 21 years and 25% (26 students) were from 21 to 30 years of age.

The students were divided into 35 three-member teams who worked together to develop the software. In this quasi-experiment, a subject is a three-member development team. These teams were formed at random and their members were blind to the quasi-experimental conditions and hypotheses. Teams used a common agile methodology (an adaptation of XP [6] to the academic environment) and the C programming language. The students were novice XP programmers.

### 3.3. Development characterization

All the participant teams were set the same project. The project was to design and implement an averagely complex distributed management software system for the Higher Technical School of Computing's library from handed out user's stories (software requirements). This way the behaviour of the teams, as well as the quality of the software product, was comparable. Development took place throughout the whole semester and professors acted as users. The size of the resulting products ranged from 87.15 to 130 adjusted function points.

Students were asked to use the XP agile process to design and develop software for the problem they had been set. The way the teams worked conformed to the agile model and its key features:

- The size of the teams conforms to agile development stipulations.
- The important point is the work done by the team not the roles performed by its members.
- Work is organized according to brief guidelines established by each team. This improves decision making and conflict management within the team.
- Work is improved through an iterative process and successive refinements.
- Students try to develop simple and useful items and do not try to anticipate needs.
- There are partial deliverables that provide feedback to the teams and prevent errors at the end of development.

We established what techniques were to be used before starting development. Apart from using the XP development process, students were told to apply construction testing, following Beck's instructions [5]. We also asked students to follow XP practices related to design (refactoring, simple design), development (trio programming and real-time code inspection, coding standards) and testing (test-driven development) defined for the course at the programming web site. Finally, they had to hand in a report briefly describing their project work and attaching the implemented code.

### 3.4. Instrumentation and data collection procedure

The instruments designed to execute the quasi-experiment and the pre-training session activity were as follows:

- *Definition of development methods*: this materialized in the set of XP practices to be executed during the experiment. They were

defined according to the original XP method, where the practices to be applied were confined to design-, construction- and testing-related practices. Planning-related practices were left out. We also established that unit tests were to be defined manually using test cases, because participants were not familiar with automated unit testing frameworks. Nevertheless, they were to be executed at the times proposed by XP, i.e. before design and coding.

- Training slides: for training in XP.
- Requirements document: this contains the set of story cards (requirements) for developing the system on a scale that was feasible in the time available for the experiment, and reasoning on how it was formulated.
- Template for specifying test cases.
- Lists of participant students and lists of teams with session observation columns.
- Process guide: document providing participants with guidance during the experiment to assure that they implement the established practices in the correct order.

Six measurement instruments were used:

1. Spanish version of the NEO FFI test to measure team member personality [15].
2. Gross Cohesion Questionnaire to measure team member cohesion [44].
3. Intragroup Conflict questionnaire to measure the type of team conflict (both task conflict and social conflict) [27].
4. van der Vegt et al.'s questionnaire [49] to measure task interdependency.
5. Molleman's questionnaire [38] to measure task autonomy.
6. Gladstein's questionnaire [22] to measure team satisfaction.

All these questionnaires were anonymized. The teams were identified by a team number and the participants by a team member number (1, 2 or 3). This subject and participant identification was the same for all the questionnaires at each measurement time. Information was gathered by means of measurements taken before, during and after project development. The questionnaires were handed out and collected back in on the dates established previously for each phase of the quasi-experiment. In actual fact, when the students handed in the questionnaires, we looked through them to check that they had answered all the questions.

To do this, measurements were taken before the project (NEO FFI personality test), during the project (conflict, cohesion) and after the project (autonomy, interdependency and satisfaction). *Before* refers to when the project kicks off, the time during which the teams are being set up, but no team work has yet been done. This is the period when the NEO FFI personality test was handed out and collected back in. *During* is the period that, according to estimated project time, it takes to complete 45% of the development project (week 11) when the teams are working on project development. This is the period when the questionnaires concerning the team processes possibly developing within the team, *cohesion* and *conflict*, were handed out and collected back in. Finally, *after* is, according to estimated project time, when the project is 95% complete (week 23) just before project development comes to an end. This period coincides with the end of the course and is when the developed software is delivered. The questionnaires handed out at this point measure the task characteristics (*interdependency* and *autonomy*) and the *satisfaction* of the teams after their team work.

Participants should answer all the questionnaire questions and take care to rate the content of each statement accurately and honestly. The responses given by each evaluated person to the NEO FFI test questionnaire were processed as follows [15]: correction and scoring of the test; elaboration of a profile chart; profile analysis.

The profile charts were interpreted and findings derived by a professor of social psychology from the Universidad Autónoma de Madrid's School of Psychology in Spain, who helped with this research. Each questionnaire was processed. The analysis of the social desirability index provided by the NEO FFI test detected response bias and faking in some cases. These respondents were removed from the sample. A total of three people were removed. One whole team was considered missing experimental data and was not processed in the quasi-experiment. This way we aimed to demonstrate the relevance of the data provided by the questionnaire.

The final quality of the software products output by the teams was evaluated when all the projects had been completed, although each of the deliverables submitted by the teams throughout the course was evaluated separately.

### 3.5. Quasi-experiment validity

In quasi-experiments, as in experiments, you have to evaluate both internal validity and external validity. In quasi-experimental designs, there is usually much less certainty about the cause of variations in the dependent variable than in experiments. Therefore, there are a number of potential threats to the investigation's internal validity that should be taken into account.

Before running the quasi-experiment, we identified seven main internal validity threats:

1. *Team size.* According to [3], smaller teams, from three to six people, are more satisfactory; it is easier to create bonds between their members; there is less competition, and they are likely to offer a more favourable work climate. Team size could therefore have an influence on team cohesion. To prevent this threat, all the teams were limited to a size of three.
2. *Participant attendance.* It is not possible to guarantee that all the teams attend the classroom team practice sessions. The action we took was to make it compulsory for students to attend classes, and we had attendance reflected in the grading of the developed software product (see participation factor in the weighted grade formula for the software product quality variable). The measure was successful and recorded attendance was high.
3. *Process conformance.* It is not possible to guarantee that the students do all the specified tasks in the required order or distribute the work properly. This was mitigated by allocating two professors to every 60 students. These professors were responsible for monitoring project development. Specifically, the professors noted down the tasks performed and the order followed in the session observations column on the team lists. If the teams were not doing the right tasks or were doing the tasks in the wrong order, they were told so and the professors went through the process guide supplied to participants with them. Process conformance was generally satisfactory, as the recorded incidents were very low.
4. *Potential impact of XP on team processes.* Agile methods are known to facilitate and motivate team building, things that were not previously addressed by non-agile methods. One might think that the usage of the agile XP process could influence the cohesion and conflict and the project results. But this was mitigated, on the one hand, by having all the teams follow a common methodology involving adapting XP to an academic environment. This was monitored and controlled as explained above. On the other hand, the XP practices were confined to those programmer practices related to technical design, construction and testing activities. Coaching practices, such as participative sessions aimed at achieving accepted conscien-

tiousness and cover for the team, or conveying a faithful reflection of the state of affairs [7], which could potentially influence team processes and consequently the results, were not applied.

5. *Knowledge of XP and development experience.* Although all the students are novice XP programmers, not all teams have the same knowledge of XP and the same experience in the task to be developed. This was mitigated by forming teams at random and organizing a training session on the XP methodology applied to project development.
6. *Team member motivation and competencies.* Not all the team members have the same competencies, like intrapersonal and interpersonal skills, not to mention aspects like motivation. This was mitigated by forming teams at random.
7. *Software project to be developed.* If the teams work on different problems, the teams' results might not be comparable. To prevent this threat, all the teams undertook the same project.

During the development of the quasi-experiment, we identified an unforeseen risk that was potentially harmful to the internal validity of the quasi-experiment:

8. *Questions about the problem.* The experimenters could become a factor of bias and should not therefore solve problems on a one-to-one basis. The mitigation measure was to establish an out-loud question time about requirements and then leave problem solving to the developers.

With respect to external validity, as the students participating in our study were novice XP users, the findings are confined to teams with novice members. Further studies with teams of experts would be required to be able to generalize these findings. Our study considers small, three-member, teams, and the findings can be generalized to small teams, with three to six members. To generalize the findings to any team size, studies would first have to be conducted with larger teams. Our results can be generalized to academic environments. For them to be generalized to an industrial setting, special-purpose quasi-experimental studies at software developer organizations would have to be planned and designed. However, if we do detect any positive relationships, the resulting recommendations could be applicable in the software industry. Additionally, such relationships could be used to formulate the research hypotheses to be tested in software developer organizations.

#### 4. Data and results analysis

The following statistical techniques were applied for data and results analysis:

1. Descriptive analysis of the five team member personality factors: *neuroticism*, *extraversion*, *openness*, *agreeableness* and *conscientiousness*.
2. Analysis of the correlations between the *personality factors*, task characteristics (*autonomy* and *interdependency*), team processes (*cohesion*, *task conflict* and *social conflict*), *satisfaction* and software product *quality*.
3. Regression between *satisfaction* and personality factors, *social conflict*, *task conflict*, *cohesion* and task *interdependency* and *autonomy*.

The results for each analysis are detailed in the following.

##### 4.1. Descriptive analysis of personality factors

According to the standardized Spanish version of the NEO FFI test [15], the students of the 35 teams that participated in the study have average values, that is, within the normal range, for

**Table 3**

Total descriptives of the subjects studied

|                   | Mtest | M     | SD   | Min | Max |
|-------------------|-------|-------|------|-----|-----|
| Neuroticism       | 15    | 18.11 | 4.42 | 9   | 26  |
| Extraversion      | 33    | 32.38 | 4.49 | 24  | 42  |
| Openness          | 30    | 29.43 | 4.44 | 19  | 40  |
| Agreeableness     | 33    | 27.97 | 3.72 | 18  | 33  |
| Conscientiousness | 36    | 29.76 | 4.42 | 22  | 39  |
| Autonomy          |       | 18.75 | 1.37 | 16  | 22  |
| Interdependency   |       | 26.34 | 3.12 | 17  | 33  |
| Satisfaction      |       | 12.38 | 2.13 | 5   | 15  |
| Conflict          |       | 13.50 | 2.77 | 9   | 19  |
| Cohesion          |       | 25.32 | 2.30 | 21  | 31  |

the extraversion (32.38) and openness (29.43) factors, as shown in Table 3.

As we can see from Table 3, however, the mean score for *neuroticism* (18.11) is slightly higher than the test average, whereas the mean scores for the *agreeableness* and *conscientiousness* factors (27.97 and 29.76, respectively) are lower than the standard scores defined in the test. The *MTest* indicates the mean of the Spanish version of the NEO FFI. *M* indicates the *mean* for the evaluated subjects, *SD* is the *standard deviation*, *min* represents the *minimum* and *max* is the *maximum* value.

The distribution of descriptive personality scores in Table 3 for students makes it harder to confirm hypotheses such as those found in other studies [3] that relate team *performance* positively to *conscientiousness* and *agreeableness* and negatively to *neuroticism*. Even so, it is certainly not surprising that being students they should be less conscientious and more neurotic about undertaking new projects as a team.

##### 4.2. Correlations of team factors

The initial analysis of correlations between *personality factors*, task *autonomy* and *interdependency*, team processes (*cohesion*, *task conflict* and *social conflict*) and *satisfaction* revealed some positive and significant associations (see Table 4). The coefficients of the significant correlations (99% confidence) are marked in Table 4 with ‘\*\*’, i.e. they have a significance level of 0.01, meaning that the error probability is less than 1%. The coefficients of correlation confirming positive relationships between the factors at a confidence level of 95% in Table 4 are marked with ‘\*’, that is, they have a significance level of 0.05, and the error probability is <5%.

Table 4 shows that *neuroticism* correlates negatively with *conscientiousness* ( $r = -0.636$   $p = 0.000$ ) and *extraversion* ( $r = -0.336$   $p = 0.049$ ). Similarly, there is a positive correlation in the teams between *extraversion* and another two personality factors, *openness* ( $r = 0.414$   $p = 0.014$ ) and *agreeableness* ( $r = 0.480$   $p = 0.003$ ), apart from with the team process, *cohesion* ( $r = 0.469$   $p = 0.005$ ). From these results, we can say that  $H_1$  is not fully supported, as it is accepted only for some personality factors: (a) there is a direct negative relationship between team member *neuroticism* and *conscientiousness*; (b) there is a negative relationship between team member *neuroticism* and *extraversion*; (c) there is a positive relationship between team member *extraversion* and *openness* to *experience*, and (d) there is a direct positive relationship between team member *extraversion* and *agreeableness*.

*Conflict* is divided, as already mentioned, into *social conflict* and *task conflict*. The evaluated *task conflict* has a significant negative relationship to *cohesion* ( $r = -0.405$   $p = 0.016$ ) and *satisfaction* ( $r = -0.526$   $p = 0.001$ ). Likewise, the evaluated *social conflict* has a significant negative relationship to *cohesion* ( $r = -0.480$   $p = 0.003$ ) and *satisfaction* ( $r = -0.354$   $p = 0.037$ ). This indicates that  $H_{12}$  is accepted, whereas  $H_7$  is not fully supported, as it only holds for the group process conflict. Therefore, it is accepted that there is a neg-

**Table 4**

Correlation matrix between personality factors, task characteristics, team processes, satisfaction and software quality

|                   | Neuroticism | Extraversion | Openness | Agreeableness | Conscientiousness | Social conflict | Task conflict | Cohesion | Autonomy | Interdependency | Satisfaction | Software quality |
|-------------------|-------------|--------------|----------|---------------|-------------------|-----------------|---------------|----------|----------|-----------------|--------------|------------------|
| Neuroticism       | 1           | –0.336*      | –0.112   | –0.151        | –0.636**          | –0.013          | –0.058        | –0.186   | –0.148   | –0.214          | –0.129       | –0.332           |
| Extraversion      |             | 1            | 0.414*   | 0.480**       | 0.256             | –0.181          | –0.164        | 0.469**  | 0.302    | 0.281           | 0.153        | 0.455*           |
| Openness          |             |              | 1        | 0.302         | 0.052             | 0.243           | 0.170         | –0.061   | –0.085   | 0.090           | 0.068        | 0.042            |
| Agreeableness     |             |              |          | 1             | 0.167             | –0.197          | –0.060        | 0.376*   | 0.503**  | 0.500**         | 0.334*       | 0.129            |
| Conscientiousness |             |              |          |               | 1                 | 0.025           | –0.022        | 0.261    | 0.216    | 0.476**         | 0.341*       | 0.240            |
| Social conflict   |             |              |          |               |                   | 1               | 0.637**       | –0.480** | –0.331   | –0.178          | –0.354*      | –0.139           |
| Task conflict     |             |              |          |               |                   |                 | 1             | –0.405*  | –0.215   | –0.287          | –0.526**     | –0.185           |
| Cohesion          |             |              |          |               |                   |                 |               | 1        | 0.333    | 0.421*          | 0.294        | 0.227            |
| Autonomy          |             |              |          |               |                   |                 |               |          | 1        | 0.599**         | 0.471**      | –0.043           |
| Interdependency   |             |              |          |               |                   |                 |               |          |          | 1               | 0.797**      | 0.268            |
| Satisfaction      |             |              |          |               |                   |                 |               |          |          |                 | 1            | 0.181            |
| Software quality  |             |              |          |               |                   |                 |               |          |          |                 |              | 1                |

\*  $p < 0.05$ .\*\*  $p < 0.01$ .

ative relationship between both *task conflict* and *social conflict* within the team and *satisfaction* of the software development team. Also there is a direct positive relationship between social conflict and task conflict ( $r = 0.637$   $p = 0.000$ ), confirming that some degree of both aspects of intragroup conflict always exists simultaneously [22].

There is a positive linear relationship between students' mean *satisfaction* and mean team personality scores for *agreeableness* and *conscientiousness* ( $r = 0.334$   $p = 0.050$  and  $r = 0.341$   $p = 0.045$ ). This means that  $H_3$  is partially accepted. Specifically, we can say that: (a) there is a positive relationship between the personality factor *agreeableness* and *satisfaction* in software development teams; and (b) there is a positive relationship between the personality factor *conscientiousness* and *satisfaction* in software development teams.

*Satisfaction* also correlates positively with task *interdependency* ( $r = 0.797$   $p = 0.000$ ) and *autonomy* ( $r = 0.471$   $p = 0.004$ ). From these results, we can say that  $H_5$  is accepted in our experimental study. In the practical assignment groups, *interdependency* is related to *autonomy* ( $r = 0.599$   $p = 0.000$ ), *agreeableness* ( $r = 0.500$   $p = 0.002$ ), *conscientiousness* ( $r = 0.476$   $p = 0.004$ ) and *cohesion* ( $r = 0.421$   $p = 0.012$ ). From these results, we can say that our study accepts  $H_{11}$ . On the other hand,  $H_8$  is only partially supported, as the following two relationships are substantiated: (a) there is a direct positive relationship between personality factor *agreeableness* and *interdependency*; and (b) there is a direct positive relationship between the personality factor *conscientiousness* and *interdependency*. The same can be said about  $H_{10}$ , which again is only partially supported, as there is found to be a positive relationship between *interdependency* and the team process *cohesion*.

No significant relationships were found between software *quality* and other evaluated factors like *autonomy*, *interdependency*, *satisfaction*, *task conflict*, *social conflict* or *cohesion*. Therefore, hypotheses  $H_4$  and  $H_6$  are rejected. Note, however, that all the evaluated teams exhibit a significant correlation between the personality factor *extraversion* and *software quality* ( $r = 0.455$   $p = 0.038$ ), as shown in Table 4. So,  $H_2$  is accepted for this personality factor only, and we can say that there is a positive relationship between the personality factor *extraversion* and *software quality*.

Analysing the individual performance of the 105 components (theory grade) against team performance, we find that there is a significant relationship ( $r = 0.201$   $p = 0.040$ ), as shown in Table 5. Therefore, average levels of extraversion in the team can be said, on the one hand, to improve interpersonal relationships (*agreeableness*) and development through social interaction (*cohesion*). This team participation improves the ties between its members, strengthening team vision and cohesion. So,  $H_9$  is accepted for only two personality factors, *extraversion* and *agreeableness* and for one

**Table 5**

Correlation between individual and group performance

|                        | Group performance |
|------------------------|-------------------|
| Individual performance | 0.201*            |

\*  $p < 0.05$ .

group factor, team *cohesion*, as there is a direct positive relationship between *extraversion* and *cohesion* and between *agreeableness* and *cohesion*.

On the other hand, high levels of *neuroticism* in the team have a negative impact on how thoroughly the team develops software (*conscientiousness*) and lead team members to shy away from interaction and communication (*extraversion*).

As mentioned earlier, the more *conflict* there is found to be in the team, the less *cohesion* there is between its members and the less satisfied they are with their jobs. But, this team *satisfaction* increases the pleasanter and more attentive team members are to each other (*agreeableness*) and the more thoroughly they do their job (*conscientiousness*).

In turn, *satisfaction* is very much related to the agile method applied by team members to develop software. In this case, we have used a method characterized by non-systematic development, very short-term schedules and implementations. This method has need of close interdependency between team members and team decision making on how to do the work (e.g. assignation of people to roles, documentation, application of processes, etc.). This is only possible if the team is generous and is committed to and is actively involved in applying the processes, of which it has a unanimous vision accepted by all team members. To assure successful development, the teams applying this development method must be highly communicative and participatory. Averagely extraverted, social and participatory teams, we found, were significantly related to better software.

#### 4.3. Linear regression of satisfaction

We built a backward model for regression, in which we entered all the variables and then eliminated them one by one based on the output criteria (lowest absolute value of  $R^2$ ; criterion of  $F \geq 0.1$ ). This produced the model shown in Table 6.

Note firstly that we applied a two-step procedure, arriving at a corrected coefficient of determination or correlation  $R^2$  of 0.733 (out of 1). In other words, the model predicts or explains that degree of variance of the variable *satisfaction* from task *interdependency*, *cohesion* and *task conflict*. A '–' sign in Table 6 indicates a negative relationship. Interestingly, the assessed social conflict plays no role in explaining the model.

**Table 6**  
Regression model for satisfaction

|                 | Non-standardized regression coefficients | Standardized regression coefficients |
|-----------------|------------------------------------------|--------------------------------------|
|                 | B                                        | Beta                                 |
| (Constants)     | 8.013                                    |                                      |
| Interdependency | 0.521                                    | 0.765                                |
| Cohesion        | −0.168                                   | −0.182                               |
| Task conflict   | −0.674                                   | −0.380                               |

$R^2 = 0.733$ .

**Table 7**  
Correlations between satisfaction, social conflict, task conflict and interdependency

|                 | Satisfaction | Social conflict | Task conflict | Interdependency |
|-----------------|--------------|-----------------|---------------|-----------------|
| Satisfaction    | 1            | −0.354*         | −0.526**      | 0.797**         |
| Social conflict |              | 1               | 0.637**       | −0.178          |
| Task conflict   |              |                 | 1             | −0.287          |
| Interdependency |              |                 |               | 1               |

\*  $p < 0.05$ .

\*\*  $p < 0.01$ .

Of the evaluated variables, the best predictor of satisfaction is *interdependency* with a beta coefficient equal to 0.765. Precisely, as we are dealing with small teams carrying out activities that they have to successfully complete very quickly (very short-term product quality), *interdependency* is indispensable, as, otherwise, there would be a very big sense of frustration at not achieving the objective. This is because in agile software development all the tasks are closely interrelated and must be performed rapidly. Consequently, there are also close and dynamic mutual dependencies between development team members.

Table 7 shows how the linear regression converges with correlations between model variables, where *interdependency* and *satisfaction* correlate ( $r = 0.797$   $p = 0.000$ ). We also find that *task conflict* ( $r = -0.526$   $p = 0.001$ ) has a closer negative relationship than *social conflict* ( $r = -0.354$   $p = 0.037$ ). In other words, according to Table 7, if there is conflict between team members, it will be harder to reach agreements on how to do the task, which is logical.

## 5. Discussion

Fig. 2 is a diagram of the results of our study. Note that the personality factors with the greatest number of correlations are *agreeableness* and *extraversion*. Fig. 2 also highlights the importance of the group factor *cohesion*, because of its relationships to both the above personality factors and the task characteristics.

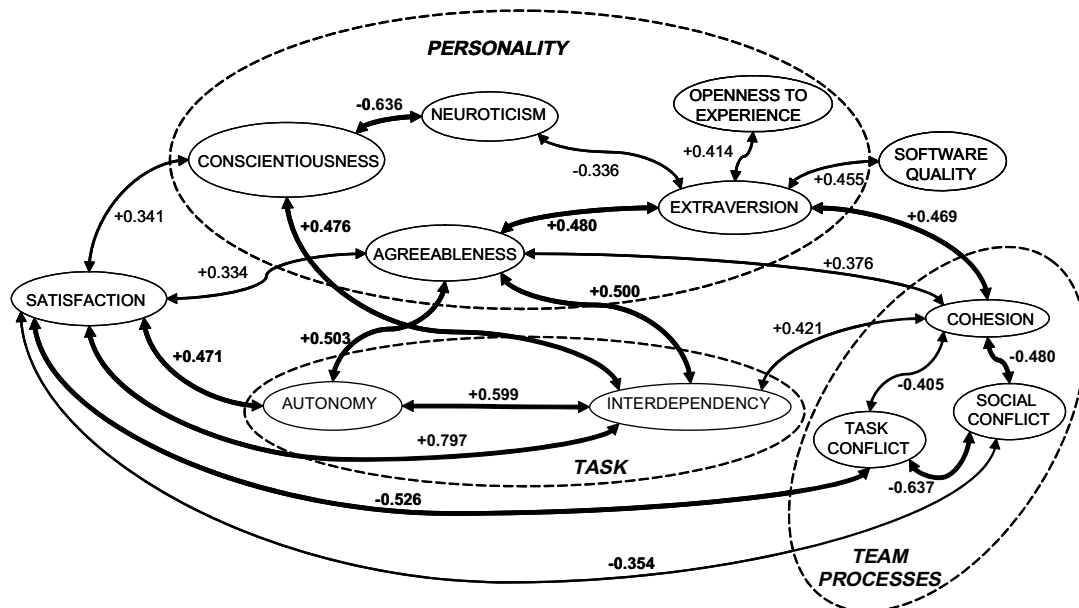
In Fig. 2, we find several relationships between different personality factors, some of which are easier to understand and interpret than others. For example, the personality factor *extraversion* is clearly related to *openness* and *agreeableness*. This is understandable bearing in mind that the relationships are friendlier and simpler in teams with high levels of extraversion, where there are no tensions among members. Also, it is reasonable to think that people will find it easier to suggest innovations on how to do their work within such a team atmosphere (openness to experience). Therefore, the quality of the work done by teams like these could possibly improve, as could, as a result, the product, in this case the developed software.

Logically speaking, high levels of *extraversion* could strengthen team *cohesion* and, under such conditions, where the team is more united, this could perhaps boost and add to the members' *openness* and *agreeableness*. These results for software development following XP, matched findings by Barrick et al. [3] for different tasks, whose key characteristic was, however, interdependency.

There are several points worth making about satisfaction (Fig. 2). We found that the most satisfied students were precisely the ones that scored higher on *agreeableness* and *conscientiousness*. In other words, the members of teams where there is a friendlier and/or more conscientious atmosphere feel more satisfied. The levels of satisfaction are also higher when the team members can decide on how to develop and organize the work they are to do.

Finally, the more *task conflict* and *social conflict* there is within the team, the less satisfied team members are with their job. Similarly, the more cohesive the team is, the less conflict there is between team members.

We found only one clear relationship in the assessed teams for software quality, and this was to *extraversion* (Fig. 2). The distribution of the personality factors among the students participating in the study differs from the NEO FFI benchmark. Whereas the stu-



**Fig. 2.** Correlations between personality, team processes, task characteristics and quality or satisfaction.

dents' scores for the factors of *extraversion* and *openness* are more or less average, *neuroticism* is slightly higher than the standard values and the scores for *conscientiousness* and *agreeableness* are both lower than the yardstick. Therefore, the profile of our sample of participants is not conducive for testing the hypotheses of our study, where a more representative sample of participants for these factors (as in the case of the analysed social psychology studies) would have been better. The results might conceivably have been clearer with greater sample variability. This would partly explain the lack of statistically significant relationships with developed software product quality, which were identified by other researchers. Even so, the established relationship matches the results obtained by Barrick and Mount [2], Barry and Stewart [4] and Barrick et al. [3]. Although these papers dealt with different task types, they all had in common that interaction among team members was high and team members shared a unanimous vision of the objectives to be achieved. These are also key characteristics of the agile development method. Therefore, teams composed of averagely extraverted people are considered to be beneficial for software product quality applying this development method.

## 6. Conclusions

On the basis of the results obtained in this quasi-experiment, we can say that the teams most *satisfied* with their job are precisely the ones that score higher on the personality factors of *agreeableness* and *conscientiousness*. The levels of *satisfaction* are also higher when the team members can decide on how to develop and organize their work. Contrariwise, the level of *satisfaction* and *cohesion* drops the greater *task conflict* is among team members. Finally, there is a significant correlation between an average score for the personality factor *extraversion* and evaluated software product quality.

Consequently, *extraversion* is significantly related to better software quality for developing software following an agile methodology. A high interaction among team members is essential for this development method. All participants could be classed as project managers and are all responsible for the success or failure of the developed product. This way, traits like sociability, talkativeness, communicativeness, affability and openness appear to be conducive to the development of high quality software, as well as to the satisfaction of the development team members.

The generalization of our findings to an industrial environment requires more studies. However, the relationships detected in this experiment are promising. These results should encourage other researchers (as well as ourselves) to continue examining the relationships between personality, team processes, task characteristics and software quality and team member satisfaction. The results of our quasi-experiment suggest that the focus should be either on the personality factor *extraversion* and software quality or on task interdependency and task conflict with team member satisfaction. A thorough examination of these relationships would enable software managers to influence team performance in two ways. The first would be by deciding how to form teams and raise software quality. The second would be by deciding to apply techniques to lower the conflict level and boost interdependency, thereby raising team satisfaction. Until new studies are conducted, all we know is that *extraversion* relates to the quality of the software that the team produces and that interdependency is positively related to satisfaction, whereas task conflict and social conflict are negatively related to satisfaction. What do not know is which causes which. Therefore, even if team managers promote these factors, there is no guarantee of the team producing high quality software or having high levels of satisfaction. Even so, it will reduce the chances of producing low quality software and having unsatisfied teams. The

practical implications of our findings are that team managers should:

- Measure the extraversion personality trait of their developers. The NEO FFI personality test is useful for this purpose.
- Form teams with average extraversion levels. This seems to reduce the chances of producing low quality software.
- Set interdependent tasks and apply work methods that boost interdependency and could have a positive influence on team member satisfaction.
- Collect data during development to find out what the levels of task and social conflict within the development team and the team member satisfaction are. To do this, use the Intragroup Conflict questionnaire and the Gladstein's questionnaire, respectively.
- Check that conflict levels do not degrade team member satisfaction.
- Take actions to prevent task and social conflict from reaching above average levels and degrading team member satisfaction and tempting them to leave the team.

This study is just a first step towards understanding the relationship between personality, team processes, task characteristics and software quality and team member satisfaction. We are replicating the quasi-experiment at several European universities to evaluate whether or not the relationships that we found hold and strengthen the proposed recommendations.

This study has a number of limitations that need to be considered to correctly interpret the results. First, the correlational evidence between the different study variables does not necessarily imply any causality. To verify these results, it would be necessary to run experimental studies analysing the effects of personality factors, team processes and task characteristics on software quality and team member satisfaction. Also it would be of interest to increase the sample size in pursuit of greater variability, run a comparative study differentiating the sample by development type (heavy/agile), use samples from software developer organizations, and carry out longitudinal research. Another line of research worth considering is to check whether teams differentiated by gender and age show different personality and behaviour characteristics and whether this has an impact on software quality and team member satisfaction.

## References

- [1] S.T. Acuña, N. Juristo, Assigning people to roles in software projects, *Softw. Pract. Exp.* 34 (2004) 675–696.
- [2] M.R. Barrick, M.K. Mount, The Big Five personality dimensions and job performance: a meta-analysis, *Pers. Psychol.* 44 (1991) 1–26.
- [3] M.R. Barrick, G.L. Stewart, M.J. Neubert, M.K. Mount, Relating member ability and personality to work-team processes and team effectiveness, *J. Appl. Psychol.* 83 (1998) 377–391.
- [4] B. Barry, G.L. Stewart, Composition, process and performance in self-managed groups: the role of personality, *J. Appl. Psychol.* 82 (1997) 62–78.
- [5] K. Beck, *Extreme Programming Explained: Embrace Change*, Addison-Wesley, Reading, MA, 1999.
- [6] K. Beck, M. Beedle, A. Cockburn, W. Cunningham, M. Fowler, et al., *Agile Manifesto*, 2001. <http://agilemanifesto.org/>.
- [7] K. Beck, W. Cunningham, A laboratory for teaching object-oriented thinking, in: *Proceedings of Object-Oriented Programming Systems, Languages and Applications (OOPSLA'89)*, SIGPLAN Notices, vol. 24(10), 1989, pp. 1–6.
- [8] J. Bentley, *Programming Pearls*, second ed., Addison-Wesley, Reading, MA, 2000.
- [9] B.W. Boehm, C. Abts, W.A. Brown, S. Chulani, B.K. Clark, E. Horowitz, R. Madachy, D.J. Reifer, B. Steece, *Software Cost Estimation with COCOMO II*, Prentice Hall PTR, Upper Saddle River, 2000.
- [10] R.P. Bostrom, K.M. Kaiser, Personality differences within systems project teams: implications for designing solving centers, in: *Proceedings of the Eighteenth Annual ACM SIGCPR Conference*, 1981, pp. 248–285.
- [11] J.A. Breaugh, The measurement of work autonomy, *Hum. Relat.* 38 (1985) 551–570.
- [12] G. Burdett, R.-Y. Li, A quantitative approach to the formation of workgroups, in: *Proceedings of the ACM SIGCPR Conference*, 1995, pp. 202–212.

- [13] D. Chan, Functional relations among constructs in the same content domain at different levels of analysis: a typology of composition models, *J. Appl. Psychol.* 83 (1998) 234–246.
- [14] T.D. Cook, D.T. Campbell, *Quasi-Experimentation Design and Analysis Issues for the Field Settings*, Houghton Mifflin, Boston, 1979.
- [15] P.T. Costa Jr., R.R. McCrae, *NEO Personality Inventory, Revised*, Psychological Assessment Resources, Odessa, Florida, 1992 (Spanish version, TEA Ediciones, Madrid, 2002).
- [16] B. Curtis, The psychology of programming-in-the-large: team and organizational behaviour, in: J.-M. Hoc, T.R.G. Green, R. Samurcay, D.J. Gilmore (Eds.), *Psychology of Programming*, Academic Press, London, 1990.
- [17] B. Curtis, M. Kellner, J. Over, Process modelling, *Commun. ACM* 35 (3) (1992) 75–90.
- [18] T. DeMarco, T. Lister, *Peopleware: Productive Projects and Teams*, second ed., Dorset House, New York, 1999.
- [19] T. Dybå, T. Dingsøyr, Empirical studies of agile software development: a systematic review, *Inform. Softw. Technol.* 50 (2008) 833–859.
- [20] A. English, R.L. Griffith, L.A. Steelman, Team performance, *Small Group Res.* 35 (6) (2004) 643–665.
- [21] B.K. Evans, D.G. Fischer, A hierarchical model of participatory decision-making, job autonomy, and perceived control, *Hum. Relat.* 45 (1992) 1169–1189.
- [22] D.L. Gladstein, Groups in context: a model of task group effectiveness, *Adm. Sci. Q.* 29 (1984) 499–517.
- [23] J.R. Hackman, G.R. Oldham, *Work Redesign*, Addison-Wesley, Reading, MA, 1987.
- [24] L.T. Hardiman, Personality types and software engineers, *IEEE Comput.* 30 (10) (1997) 10.
- [25] IEEE, *Guide to the Software Engineering Body of Knowledge-SWEBOK Version 2004*, IEEE Computer Society, 2004.
- [26] A. Jedlitschka, D. Pfahl, *Reporting Guidelines for Controlled Experiments in Software Engineering*, 2005. [www.iese.fraunhofer.de/network/ISERN/pub/technical\\_reports/isern-05-01.pdf](http://www.iese.fraunhofer.de/network/ISERN/pub/technical_reports/isern-05-01.pdf).
- [27] K.A. Jehn, A multimethod examination of the benefits and detriments of intragroup conflict, *Adm. Sci. Q.* 40 (1995) 256–282.
- [28] J. Katzenbach, D. Smith, *The Discipline of Teams: A Mindbook-Workbook for Delivering Small Group Performance*, John Wiley & Sons, New York, 2001.
- [29] M.I. Kellner, R.J. Madachy, D.M. Raffo, Software process simulation modelling: Why? What? How?, *J. Syst. Softw.* 46 (1999) 91–105.
- [30] D.A. Kenny, L. La Voie, Separating individual and group effects, *J. Personality Social Psychol.* 48 (1985) 339–348.
- [31] M.A. Lieberman, I.D. Yalom, M.B. Miles, *Encounter Groups: First Fact*, Basic Books, New York, 1973.
- [32] F. Macias, M. Holcombe, M. Gheorghe, A formal experiment comparing extreme programming with traditional software construction, in: *Proceedings of the Fourth Mexican International Conference on Computer Science (ENC 2003)*, 2003, pp. 73–80.
- [33] K. Mannaro, M. Melis, M. Marchesi, Empirical analysis on the satisfaction of IT employees comparing XP practices with other software development methodologies, *Proceedings of the Fifth International Conference on Extreme Programming and Agile Processes in Software Engineering Lecture Notes in Computer Science*, vol. 3092, Springer, 2004. pp. 166–174.
- [34] R.C. Martin, eXtreme Programming development through dialog, *IEEE Softw.* July–August (2000) 12–13.
- [35] C. McDowell, L. Werner, H. Bullock, J. Fernald, Pair programming improves student retention, confidence, and program quality, *Commun. ACM* 49 (8) (2006) 90–95.
- [36] J.E. McGrath, *Groups: Interaction and Performance*, Prentice-Hall, Englewood Cliffs, New Jersey, 1984.
- [37] E. Molleman, A. Nauta, K.A. Jehn, Person-job fit applied to teamwork: a multi-level approach, *Small Group Res.* 35 (2004) 515–539.
- [38] E. Molleman, The modalities of self-management: The ‘must’, ‘may’, ‘can’ and ‘will’ of local decision making, *Int. J. Oper. Prod. Manage.* 20 (2000) 889–910.
- [39] E. Moore, Personality characteristics of information systems professionals, in: *Proceedings of the Conference on SIGCPR*, 1991, pp. 140–155.
- [40] G.A. Neuman, J. Wright, Team effectiveness: beyond skills and cognitive ability, *J. Appl. Psychol.* 84 (3) (1999) 376.
- [41] S.L. Pfleeger, *Software Engineering: Theory and Practice*, second ed., Prentice-Hall, New Jersey, 2001.
- [42] R.H. Rutherford, Using personality inventories to help form teams for software engineering class projects, *SIGCSE Bull.* 33 (3) (2001) 76.
- [43] I. Sommerville, T. Rodden, Human social and organisational factors in the software process, in: A. Fuggetta, A. Wolf (Eds.), *Trends in Software: Software Process*, John Wiley, New York, 1996, pp. 89–100.
- [44] J.P. Stokes, Toward an understanding of cohesion in personal change groups, *Int. J. Group. Psychother.* 33 (1983) 449–467.
- [45] J. Teague, Personality type, career preference and implications for computer science recruitment and teaching, in: *Proceedings of the Third Australasian Conference on Computer Science Education*, 1998, pp. 155–63.
- [46] B. Tuckman, Developmental sequence in small groups, *Psychol. Bull.* 63 (1965) 384–399.
- [47] R. Turley, J. Bieman, Competencies of exceptional and nonexceptional software engineers, *J. Syst. Softw.* 28 (1) (1995) 19–38.
- [48] D.A. Umphress, T.D. Hendrix, J.H. Cross, Software process in the classroom: the Capstone project experience, *IEEE Softw.* September–October (2002) 78–85.
- [49] G. van der Vegt, B. Emans, E. Van de Vliert, Affective reactions to individual task interdependence in outcome interdependence groups, *Pers. Psychol.* 54 (2001) 51–69.
- [50] A.E.M. van Vianen, C.K.W. De Dreu, Personality in team: its relationship to social cohesion and task cohesion and team performance, *Eur. J. Work Organ. Psychol.* 10 (2) (2001) 97–120.
- [51] C.A. Wellington, T. Briggs, C.D. Girard, Comparison of student experiences with plan-driven and agile methodologies, in: *Proceedings of the 35th ASEE/IEEE Frontiers in Education Conference*, 2005, pp. T3G-18–T3G-23.
- [52] K. White, R. Leifer, Information systems development success: perspectives from project team participants, *MIS Q* 10 (3) (1986) 215–223.
- [53] J. Wynkoop, D. Walz, Investigating traits of top performing software developers, *Inf. Technol. People* 13 (3) (2000) 186–195.
- [54] H.-L. Yang, J.-H. Tang, Team structure and team performance in IS development: a social network perspective, *Inf. Manage.* 41 (2004) 335–349.
- [55] A. Zakarian, A. Kusiak, Forming teams: an analytical approach, *IIE Trans.* 31 (1999) 85–97.
- [56] W. Zuser, T. Grechening, Reflecting skills and personality internally as means for team performance improvement, *Proceedings of the 16th Conference on Software Engineering Education and Training*, IEEE Computer Society, 2003.